very good in most parts => accept
final thesis: method for interviews
Last section

Structured Subsystem-Aware Feature Representations for Predictive Maintenance in Metro Systems

Emirhan Kurtulus[1]

TU Wien, Student ID (Matrikelnummer: 12243493),
e12243493@student.tuwien.ac.at

Supervisor: Univ.-Prof. Dr.-Ing. habil. Fazel Ansari, TU Wien
Content Support: Dipl.-Ing. Andreas Steiner, B.Sc.

Abstract. Predictive maintenance has become an essential approach for improving the reliability and efficiency of complex industrial systems. However, existing approaches typically process sensor variables as a single high-dimensional feature space and provide limited methodologies for systematically constructing subsystem-aware feature representations from recurring anomaly patterns. This thesis develops a methodology for constructing subsystem-aware feature representations from recurring anomaly patterns extracted from MetroAT anomaly windows through clustering, pattern analysis, and expert verification. The resulting representations are evaluated against conventional full-feature representations under identical experimental conditions using predictive maintenance models. The expected contribution is a systematic methodology that improves the robustness, interpretability, and reliability of predictive maintenance for metro systems.

Keywords: Predictive Maintenance · Subsystem-Aware Feature Representation · Anomaly Detection · Industrial Time Series · Metro Systems

1 Introduction

The increasing digitalization of industrial systems within the context of Industry 4.0 has enabled the continuous collection of high-frequency sensor data from complex technical infrastructures. This development has accelerated the adoption of predictive maintenance, which uses condition-monitoring data to anticipate failures before they occur, thereby improving system reliability, reducing maintenance costs, and minimizing unplanned downtime [1,6]. Predictive maintenance is particularly important in railway and metro systems, where equipment failures directly affect operational efficiency, service availability, and passenger safety [10]. Modern metro systems continuously generate multivariate sensor data describing the behaviour of interconnected technical subsystems, making machine learning and deep learning increasingly important for supporting data-driven maintenance decisions [1,8]. Despite these advances, reliable predictive

maintenance under realistic operational conditions remains challenging. This motivates the investigation of how sensor information is represented and organized, leading to the specific research problem addressed in this thesis.

2 Research Problem

Existing predictive maintenance approaches commonly process sensor variables as a single high-dimensional input space without explicitly considering subsystem-level functional relationships [1]. In complex metro systems, however, failures typically originate within specific operational subsystems such as braking, air production, or leveling systems. Ignoring these functional structures may reduce signal coherence, increase irrelevant information, and limit predictive robustness and interpretability [10].

The MetroAT benchmark dataset provides a realistic environment for evaluating predictive maintenance methods using subsystem-level sensor measurements and aligned service-affecting failure records [9]. Previous evaluations on MetroAT demonstrate that both classical machine learning and deep learning models experience limited robustness and high false-alarm rates under realistic temporal evaluation settings [10]. These limitations reduce the reliability of predictive maintenance models in operational deployment.

Although recurring anomaly patterns contain valuable information about subsystem degradation, existing approaches rarely use these patterns to systematically construct subsystem-aware feature representations. Consequently, there remains a lack of methodologies for constructing subsystem-aware feature representations from recurring anomaly patterns and evaluating their influence on predictive maintenance performance compared with conventional full-feature representations under identical experimental conditions[8,1].

3 Goals and Expected Results

The overall goal of this thesis is to develop a methodology for constructing subsystem-aware feature representations from recurring anomaly patterns extracted from MetroAT anomaly windows through clustering, pattern analysis, and expert verification. The resulting feature representations will be used to construct subsystem-specific datasets and evaluated against conventional full-feature representations using the MetroAT benchmark dataset [9].

The expected scientific contribution is a systematic methodology for constructing subsystem-aware feature representations rather than a new predictive maintenance model. Its effectiveness will be assessed by comparing predictive performance, robustness, generalization, and false-alarm behaviour under identical experimental conditions. Improvements will be evaluated using Precision, Recall, F1-score, and False Alarm Rate (FAR), while keeping the model architectures, hyperparameters, temporal train-test splits, and training procedures constant to isolate the effect of the feature representation strategy [10,2].

4 Research Questions

RQ1. To what extent can subsystem-aware feature representations constructed from recurring anomaly patterns extracted from MetroAT anomaly windows produce subsystem structures that are verified by domain experts?

This question will be investigated by identifying recurring anomaly patterns through clustering and pattern analysis, constructing subsystem-aware feature representations, and verifying the resulting subsystem structures with domain experts. The outcome will be assessed through the degree to which the constructed subsystem structures are verified by domain experts and the resulting subsystem-specific datasets.

RQ2. To what extent do subsystem-aware feature representations improve predictive maintenance performance compared with conventional full-feature representations under identical experimental conditions?

This question will be investigated by comparing subsystem-aware and conventional full-feature representations using identical model architectures, temporal train-test splits, hyperparameter settings, and training procedures. Performance will be evaluated using Precision, Recall, F1-score, and False Alarm Rate (FAR).

5 Research Methodology

This thesis follows an empirical quantitative research design consisting of two consecutive phases. The MetroAT benchmark dataset, containing multivariate sensor measurements and aligned service-affecting failure records from an operational metro system, serves as the experimental basis [9].

The first phase addresses RQ1 by constructing subsystem-aware feature representations. Recurring anomaly patterns will be identified from predefined MetroAT anomaly windows using clustering and temporal pattern analysis[8,5]. Based on these patterns, subsystem-aware feature representations will be constructed and subsequently reviewed and verified together with domain experts from Wiener Linien. The expert-verified feature representations will then be used to generate subsystem-specific datasets.

The second phase addresses RQ2 by evaluating the constructed feature representations. Selected temporal prediction models, including Random Forest and recurrent deep learning models, will be trained using both subsystem-aware and conventional full-feature representations. To isolate the effect of the feature representation strategy, identical preprocessing, temporal train-test splits, model architectures, hyperparameter settings, and training procedures will be maintained across all experiments [10,1,7,3].

Performance will be evaluated using Precision, Recall, F1-score, and False Alarm Rate (FAR). Subsystem-specific results will additionally be analyzed to assess robustness, generalization, and the influence of structured feature representations under realistic operational conditions [2]. Reproducibility will be

ensured through consistent preprocessing and standardized experimental settings.

6 State of the Art

Predictive maintenance has become a well-established research area through advances in machine learning and deep learning for industrial time-series analysis[1,6,3,7]. Existing studies have demonstrated the effectiveness of temporal models for failure prediction, while recent work increasingly emphasizes the importance of integrating domain knowledge to improve model interpretability and robustness [8,4]. The MetroAT benchmark provides a realistic evaluation framework for predictive maintenance using operational metro data and aligned failure records, highlighting the challenges of achieving robust performance under practical conditions [9,10]. Despite these developments, current research provides limited methodologies for systematically constructing subsystem-aware feature representations from recurring anomaly patterns. This thesis addresses this gap by developing and evaluating such a methodology[10].

7 Timeline

Months 1–2 focus on familiarization with the literature, analysis of the MetroAT benchmark dataset, and identification of recurring anomaly patterns from predefined anomaly windows using clustering and pattern analysis. During this phase, subsystem-aware feature representations will be constructed and verified with domain experts.

Months 3–4 are dedicated to generating subsystem-specific datasets and conducting predictive maintenance experiments using both subsystem-aware and conventional full-feature representations under identical experimental conditions.

Months 5–6 focus on analyzing and interpreting the experimental results, comparing both representation strategies, documenting the proposed methodology, and completing the thesis writing, revision, and final submission.

8 Relevance ~~of the Research~~

This research contributes to predictive maintenance by providing a systematic methodology for constructing subsystem-aware feature representations from recurring anomaly patterns in industrial time-series data. Scientifically, it addresses a gap in the structured integration of anomaly patterns and domain knowledge into feature representation design. Practically, the methodology aims to improve the robustness, interpretability, and reliability of predictive maintenance systems for metro operations, supporting earlier failure detection, reduced false alarms, and more informed maintenance planning[10,2].

References

1. Carvalho, T.P., Soares, F.A.A.M.N., Vita, R., da P. Francisco, R., Basto, J.P., Alcalá, S.G.S.: A systematic literature review of machine learning methods applied to predictive maintenance. Computers Industrial Engineering **137**, 106024 (2019). https://doi.org/https://doi.org/10.1016/j.cie.2019.106024, https://www.sciencedirect.com/science/article/pii/S0360835219304838
2. García-Méndez, S., de Arriba-Pérez, F., Leal, F., Veloso, B., Malheiro, B., Burguillo-Rial, J.C.: An explainable machine learning framework for railway predictive maintenance using data streams from the metro operator of portugal. Scientific Reports **15**(1), 27495 (2025)
3. Ismail Fawaz, H., Forestier, G., Weber, J., Idoumghar, L., Muller, P.A.: Deep learning for time series classification: a review. Data mining and knowledge discovery **33**(4), 917–963 (2019)
4. Klein, P.: Combining Expert Knowledge and Deep Learning with Case-Based Reasoning for Predictive Maintenance. Life Science and Basic Disciplines (German Language), Springer Fachmedien Wiesbaden (2025), https://books.google.at/books?id=YRJVEQAAQBAJ
5. Nedelkoski, S., Cardoso, J., Kao, O.: Anomaly detection and classification using distributed tracing and deep learning. In: 2019 19th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID). pp. 241–250 (2019). https://doi.org/10.1109/CCGRID.2019.00038
6. Nunes, P., Santos, J., Rocha, E.: Challenges in predictive maintenance – a review. CIRP Journal of Manufacturing Science and Technology **40**, 53–67 (2023). https://doi.org/https://doi.org/10.1016/j.cirpj.2022.11.004, https://www.sciencedirect.com/science/article/pii/S1755581722001742
7. Shiri, F.M., Perumal, T., Mustapha, N., Mohamed, R.: A comprehensive overview and comparative analysis on deep learning models: Cnn, rnn, lstm, gru. arXiv preprint arXiv:2305.17473 (2023)
8. Steenwinckel, B., De Paepe, D., Vanden Hautte, S., Heyvaert, P., Bentefrit, M., Moens, P., Dimou, A., Van Den Bossche, B., De Turck, F., Van Hoecke, S., Ongenae, F.: Flags: A methodology for adaptive anomaly detection and root cause analysis on sensor data streams by fusing expert knowledge with machine learning. Future Generation Computer Systems **116**, 30–48 (2021). https://doi.org/https://doi.org/10.1016/j.future.2020.10.015, https://www.sciencedirect.com/science/article/pii/S0167739X20329927
9. Steiner, A., Abdelkader, O., Helm, S., Ansari, F.: Metroat: A benchmark dataset for data-driven maintenance in metro operations. Procedia CIRP (2025), cIRP Conference on Manufacturing Systems (CIRP CMS)
10. Steiner, A., Helm, S., Abdelkader, O., Ansari, F.: Why machine learning falls short in predictive maintenance for metro trains? real-world challenges. In: Proceedings of the Annual Reliability and Maintainability Symposium (RAMS). IEEE (2026)